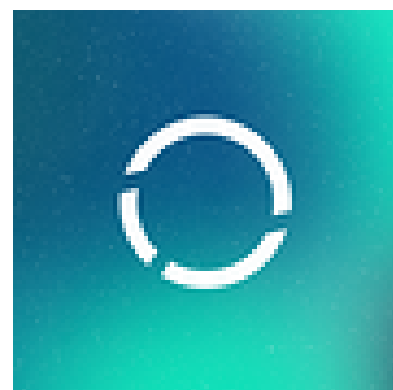




OFFZONE

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

End of Studies Project Presentation for
MASTER DATA SCIENCE & BIG DATA

ARTIFICIAL INTELLIGENCE FOR AUTOMATION: DESIGN AND IMPLEMENTATION OF AI AGENTS TO IMPROVE A SAAS COMPANY EFFICIENCY

Jury

ABDELAZIZ MARZAK (PRESIDENT)
RACHIDA AIT ABDELOUAHID (SUPERVISOR)
SOUKAINA MERZOUK (EXAMINER)

Realisation By

SALAH EDDINE
OUIRRA

Date

20/09/2025

PLAN

Context

State of the Art

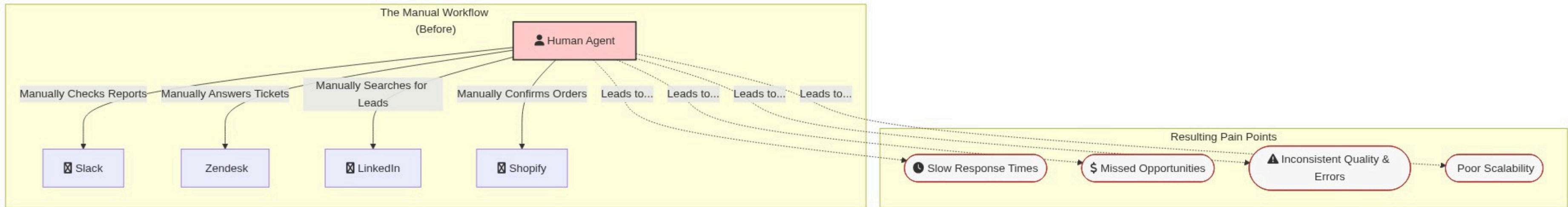
LLMs & Prompt
Engineering

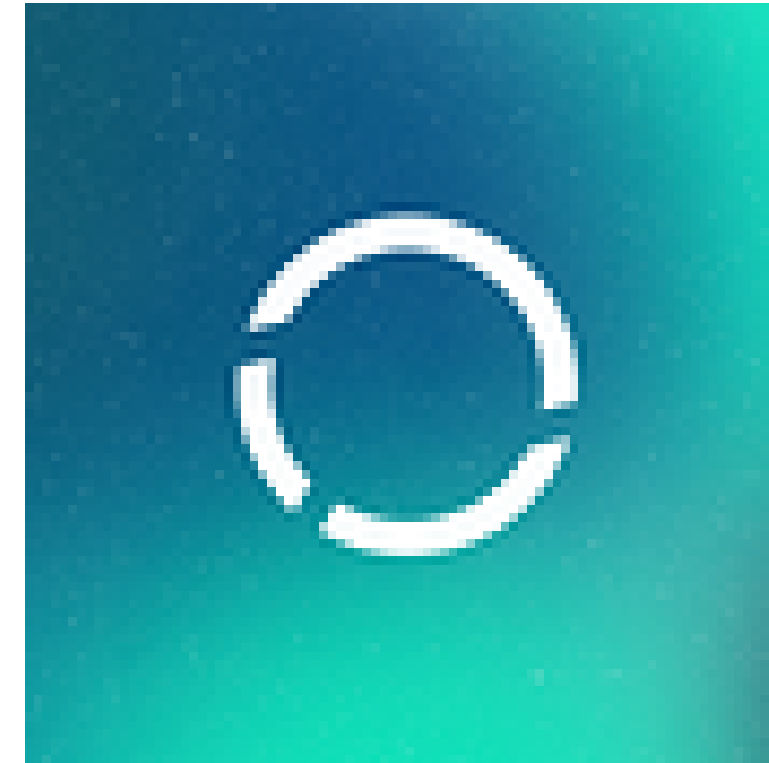
RAG & Embeddings

Implementation &
Results

CONTEXT

MANUAL FLOW



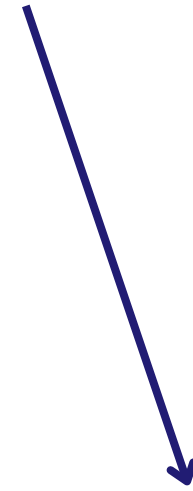


OBJECTIVE

Automate and improve our company's internal processes .

- Time saving
- Scalability
- Data centralization

AI AGENTS



STATE OF THE ART

AI AGENTS

A chain of Tools that use artificial intelligence to autonomously pursue goals and complete tasks by perceiving its environment, reasoning, planning actions

AI AGENTS

Autonomous



Defined Steps

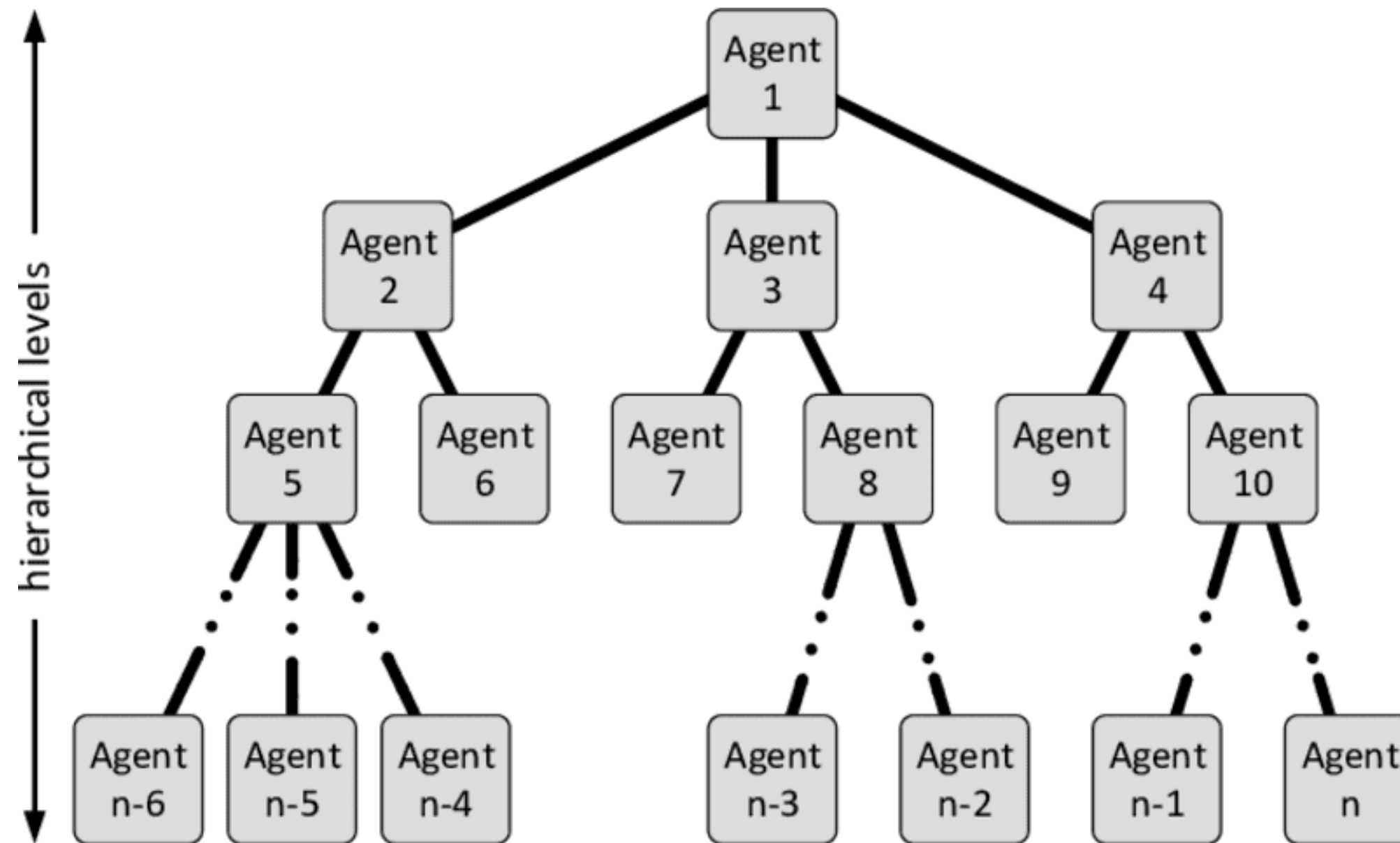
MULTI AGENTIC SYSTEMS

systems composed of multiple intelligent agents that interact and collaborate to achieve common goals.

Each agent is capable of learning, reasoning and making decisions while being able to communicate with other agents in the system.

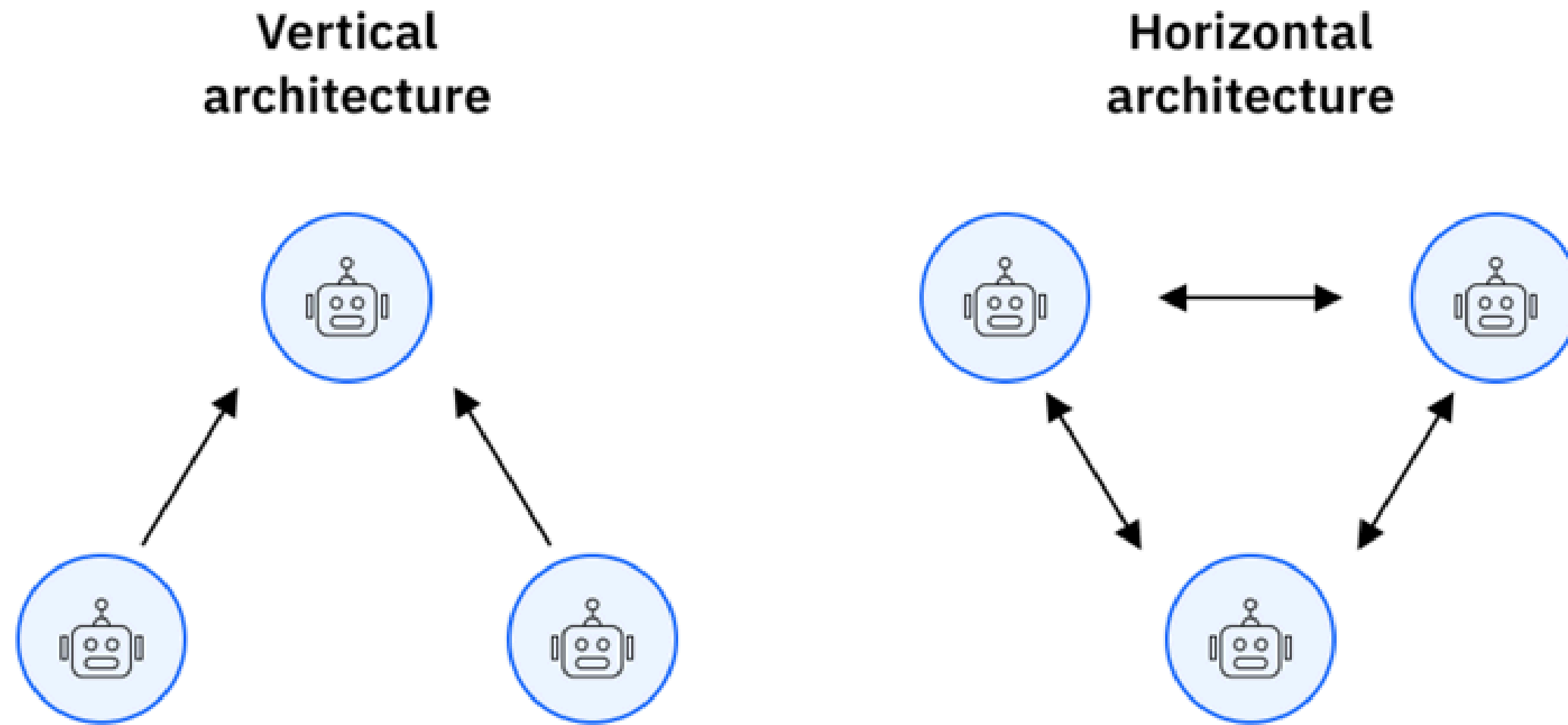
MULTI AGENTIC SYSTEMS

Architectures Hierarchic Architecture



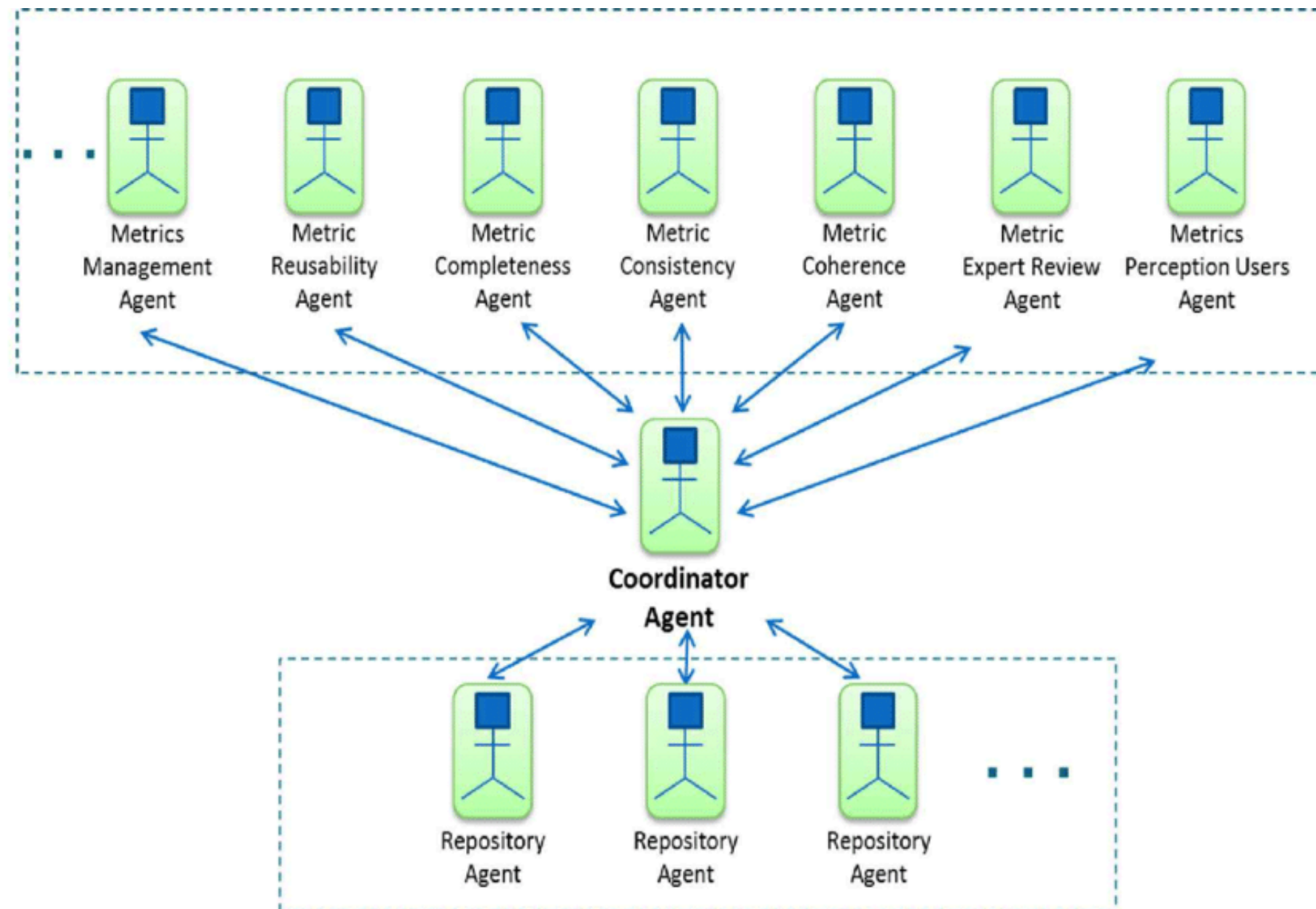
MULTI AGENTIC SYSTEMS

Architectures Horizontal Architecture



MULTI AGENTIC SYSTEMS

Architectures Hybrid Architecture



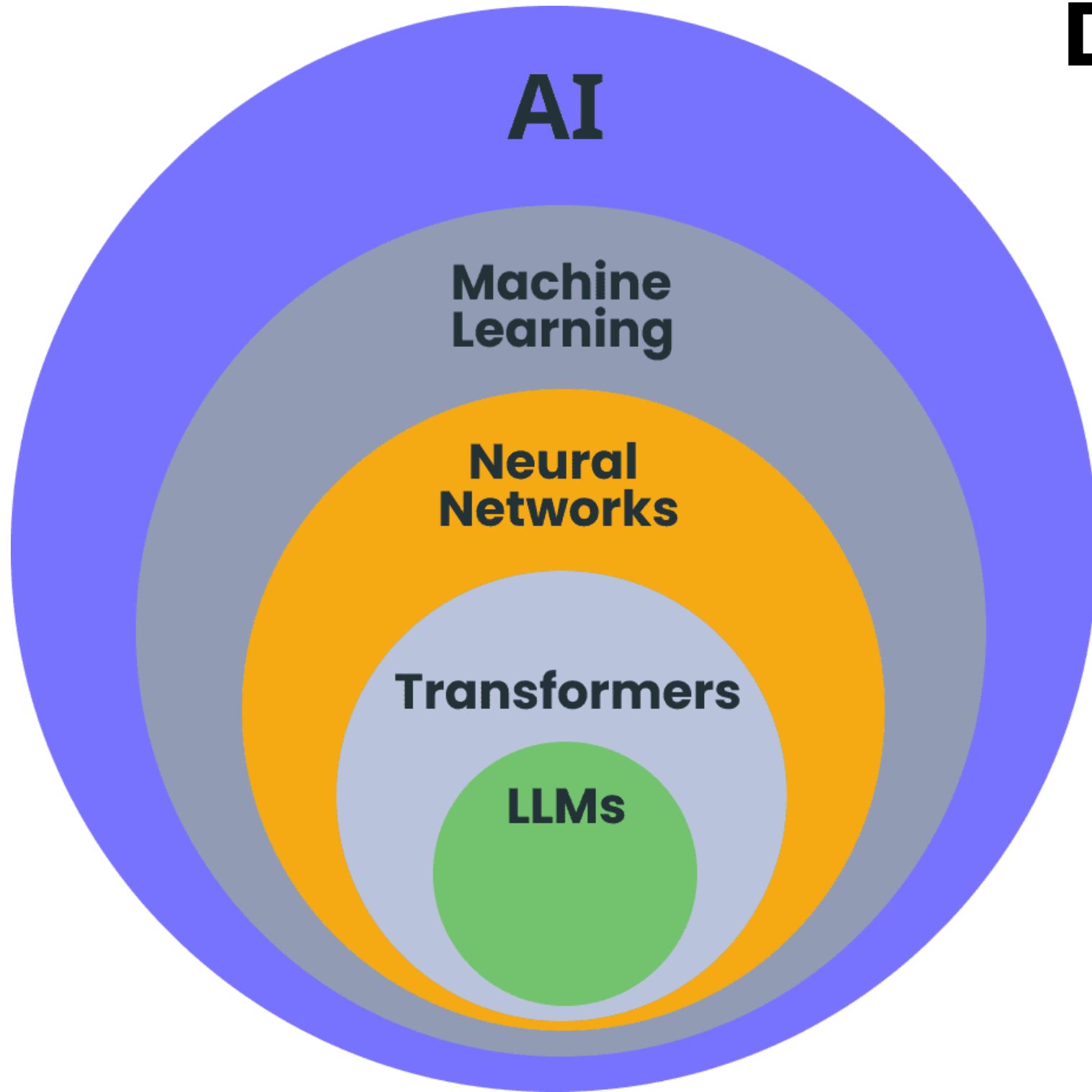
MULTI AGENTIC SYSTEMS

PAST APPROACHES

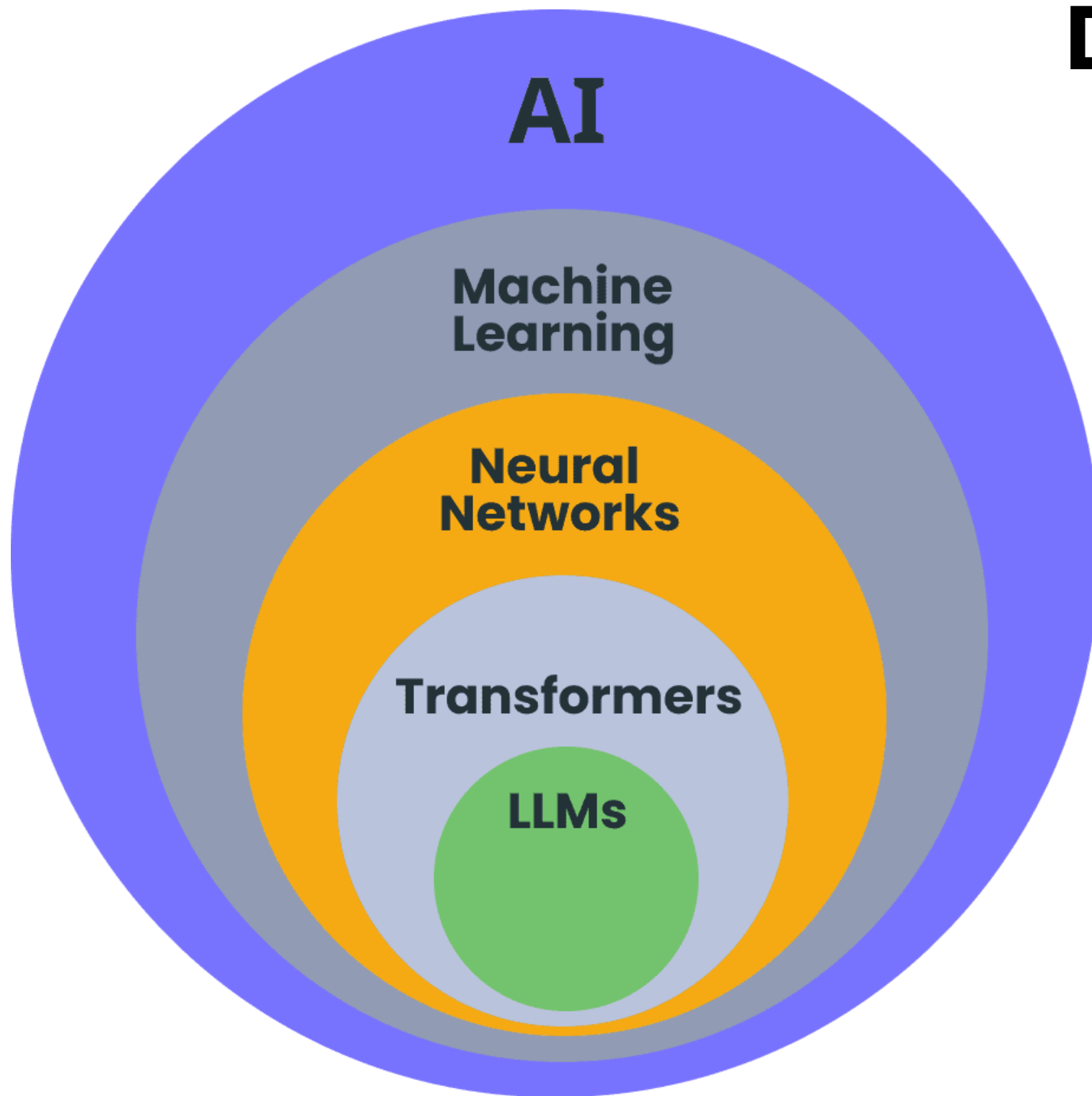
MAS	Architecture	LLM	Purpose
MetaGPT	Role-Based Simulation: Mimics a software company structure with specialized roles	Often demonstrated with GPT series	automate the entire software development process, from requirements to code generation and testing.
CAMEL	Role-Playing Dialogue: Uses "inception prompting" where two agents (an AI assistant and an AI user) are instructed to collaborate to complete a task	Often demonstrated with GPT series	study and facilitate autonomous collaboration between communicative agents
LaMAS Framework	Hybrid Architectural Patterns: Proposes multiple patterns	Multi-modal LLMs	create a broad, monetizable ecosystem for collaborative AI
Evolver Framework	Iterative Feedback Loop: A meta-system of agents	Llama 3.2-3B	autonomously improve the performance and configuration of existing MAS

LARGE LANGUAGE MODELS

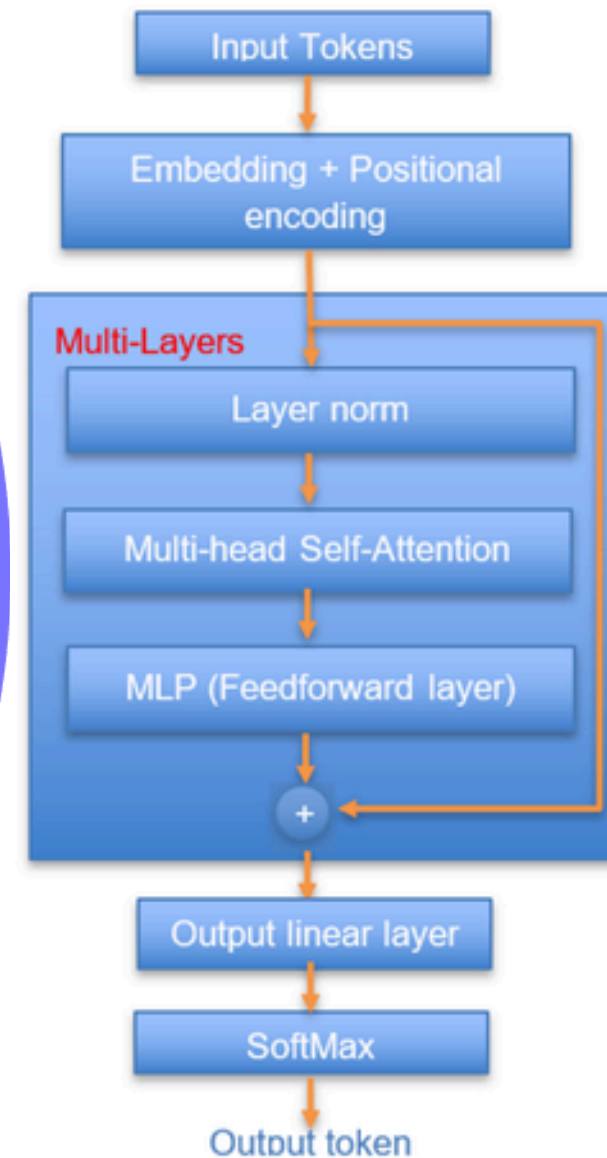
Definition



a deep neural network trained on huge text datasets to predict subsequent tokens



Definition



Parameter	Character-level	GPT-T2
Block Size (Context size)	64	1024
Number of Layers	4	12
Number of heads	4	12
Embedding Dimension	128	768
Dropout	0.2	0.2
Batch size	12	12
Vocab size	65	50257
Headen Layer dimension	4x128	4x768

Model Parameters

a deep neural network trained on huge text datasets to predict subsequent tokens

OUR CHOICE

Llama 3.1



open-weight language model family from Meta (2024)

optimized for efficiency and performance across tasks like reasoning, coding and dialogue.

Prompt Engineering



the art of designing, structuring and optimizing the instructions given to an LLM to obtain precise and context-appropriate answers

Different Types of AI Prompts



Informational Prompt

"Give a brief explanation of what GenAI is."



Creative Prompt

Compose a dramatic poem about AI-generated art.



Comparison Prompt

"Compare the capabilities of ChatGPT and GPT-4."



Persona or Role-Playing Prompt

"You're a financial advisor. What strategies would you use to help clients save for retirement?"



Instructional Prompt

"Write step-by-step instructions on how to set up a Gmail account."



Completion Prompt

"Continue this story: 'The writer stared at the blank page, willing the words to come. Suddenly, an idea struck...'"



Reasoning Prompt

"Analyze the weekly sales data from the past year and identify key factors that impact sales. Suggest strategies to optimize them."



Multi-Modal Prompt

"Examine this image of a store display and suggest enhancements to increase sales."

[Attach image of store display].



Classification Prompt

"Categorize the sentiment of these customer reviews as positive, negative, or neutral."

[Add the customer reviews to be categorized].

ng
g

Prompt Engineering

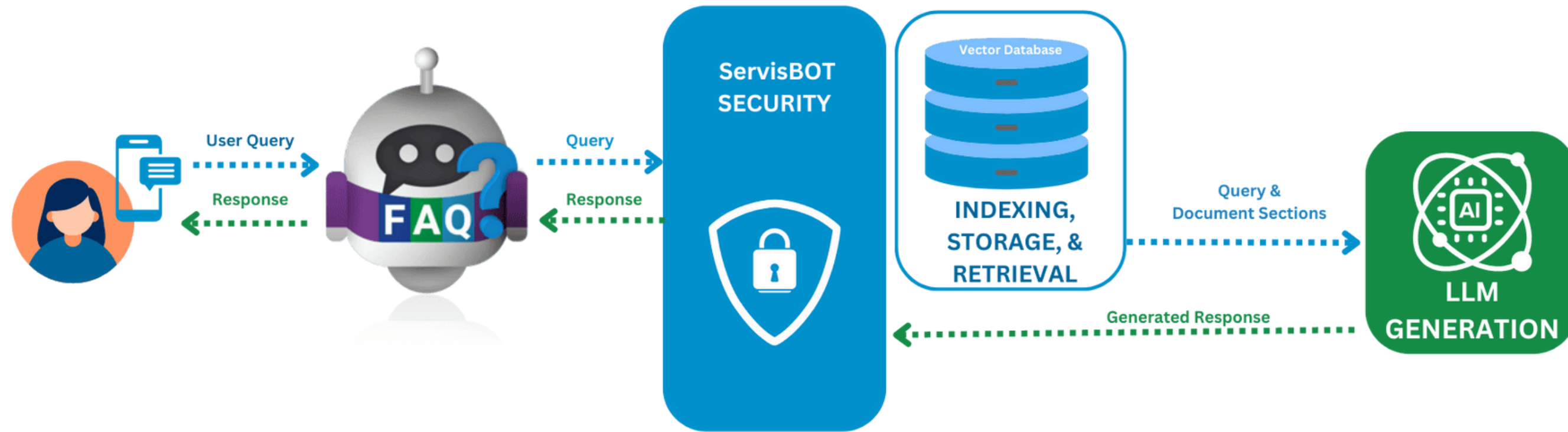
Parameter Tuning



- Temperature
- Top-p (nucleus sampling)
- Max Tokens
- Penalties

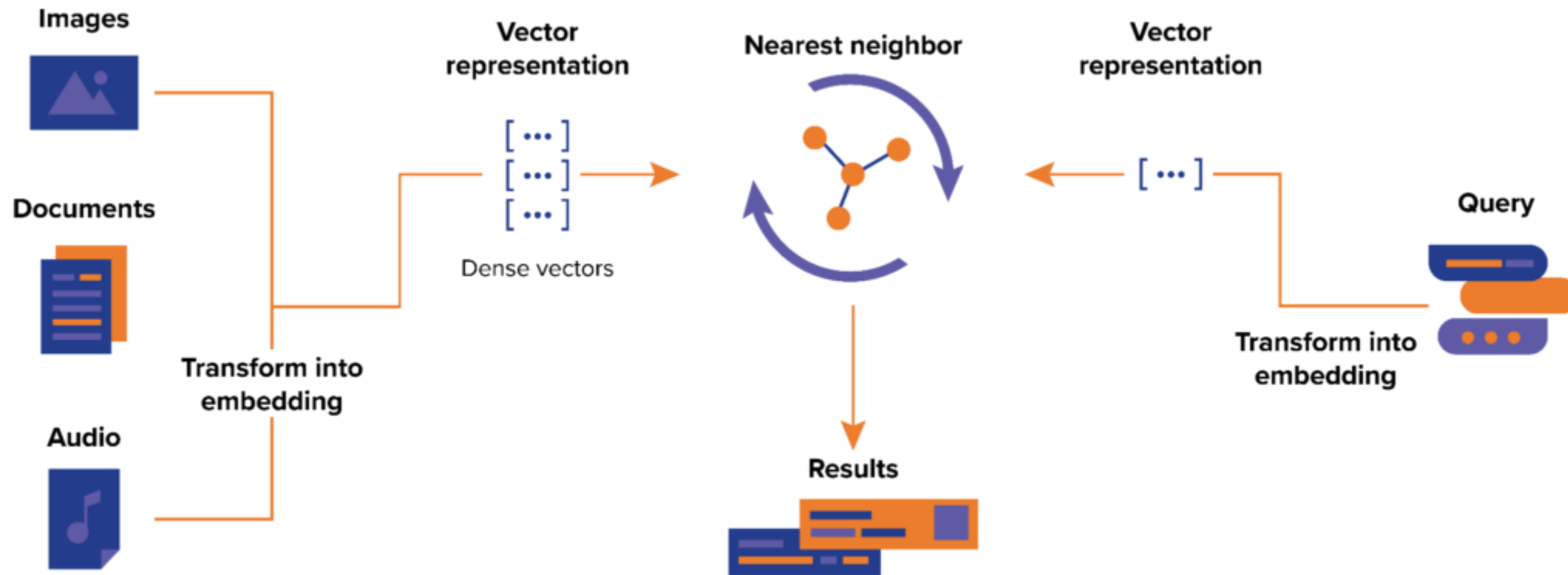
RETRIEVAL AUGMENTED GENERATION

RAG (Retrieval-Augmented Generation)



Technique that combines text generation with information retrieval to improve the quality and accuracy of generated responses.

Embeddings

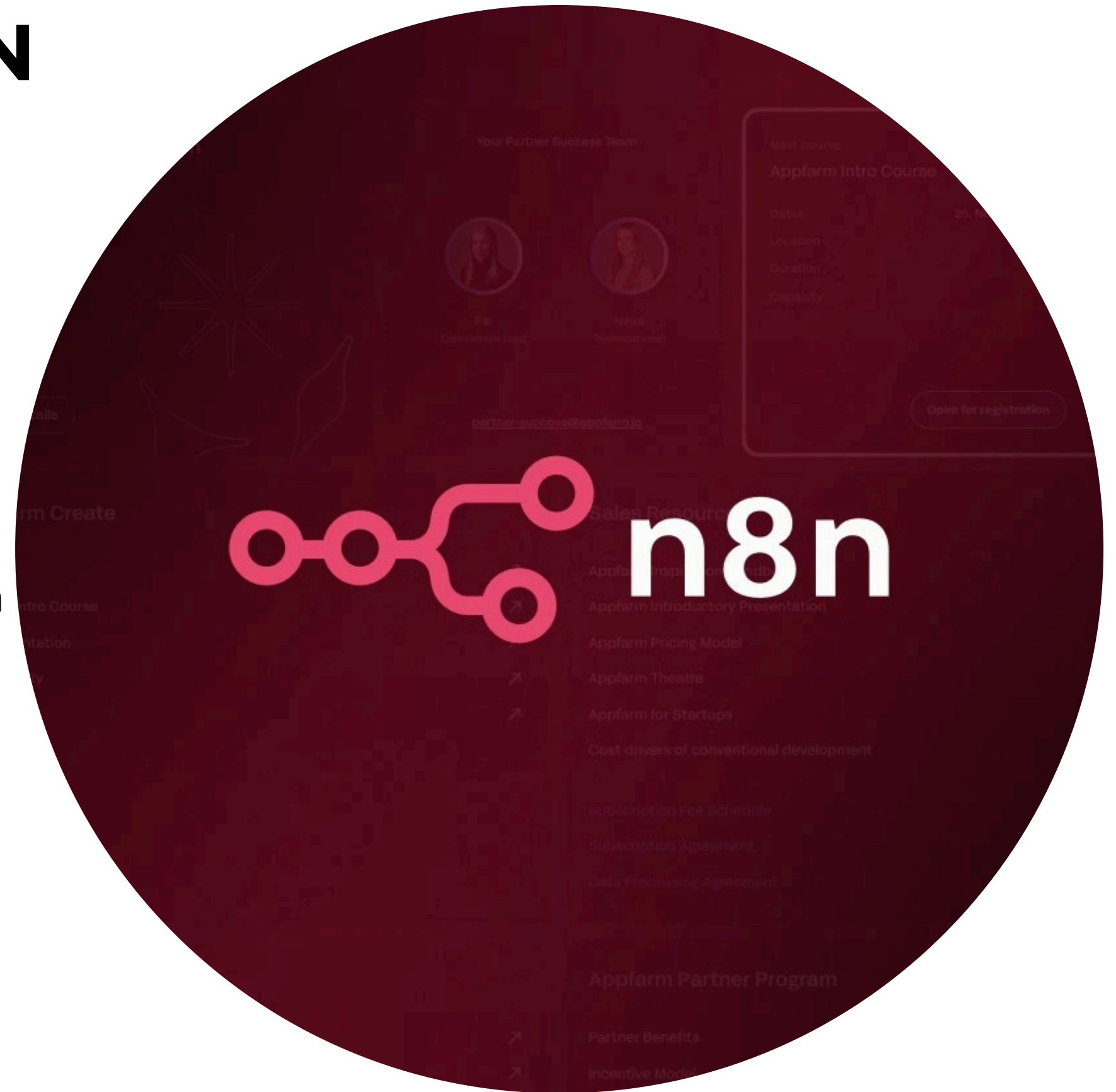


used in machine learning to represent complex data such as text or images as high-dimensional vectors.

TECHNOLOGIES

N8N

- A low-code automation platform that allows the creation of agents that visually orchestrate processes
- It allows us to run custom code (JavaScript, Python via Docker) to automate tasks with flexibility.

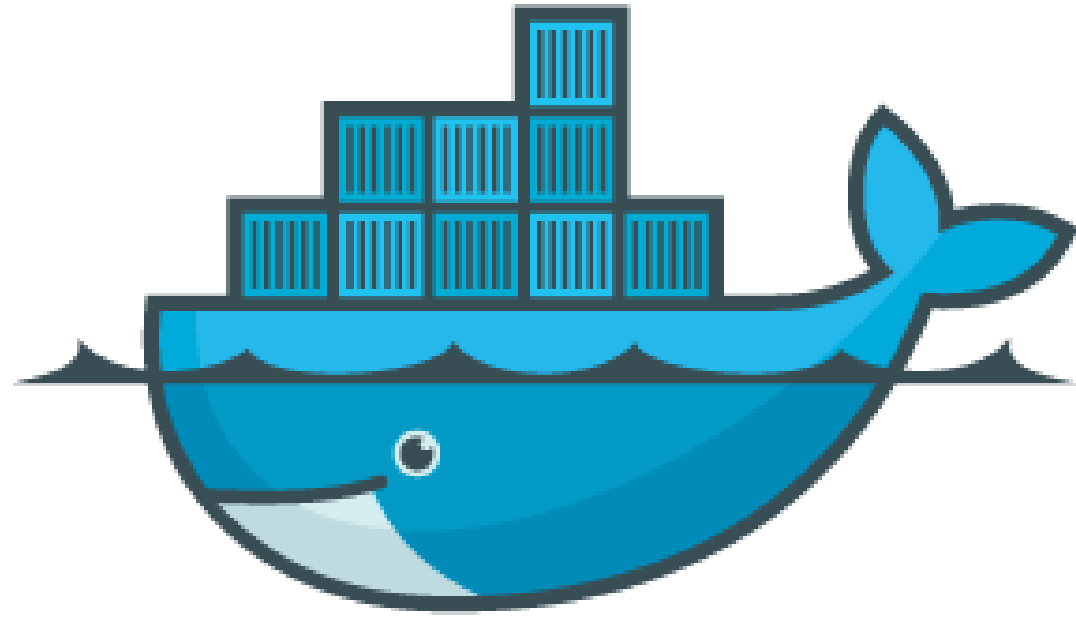


MCP servers (Model Context Protocol)



provides AI agents with access to external resources and capabilities, such as file systems, databases and communication tools

TECHNOLOGY STACK



Docker – Containerization



Ollama – Local LLM use



Qdrant – Vector Database



mongoDB



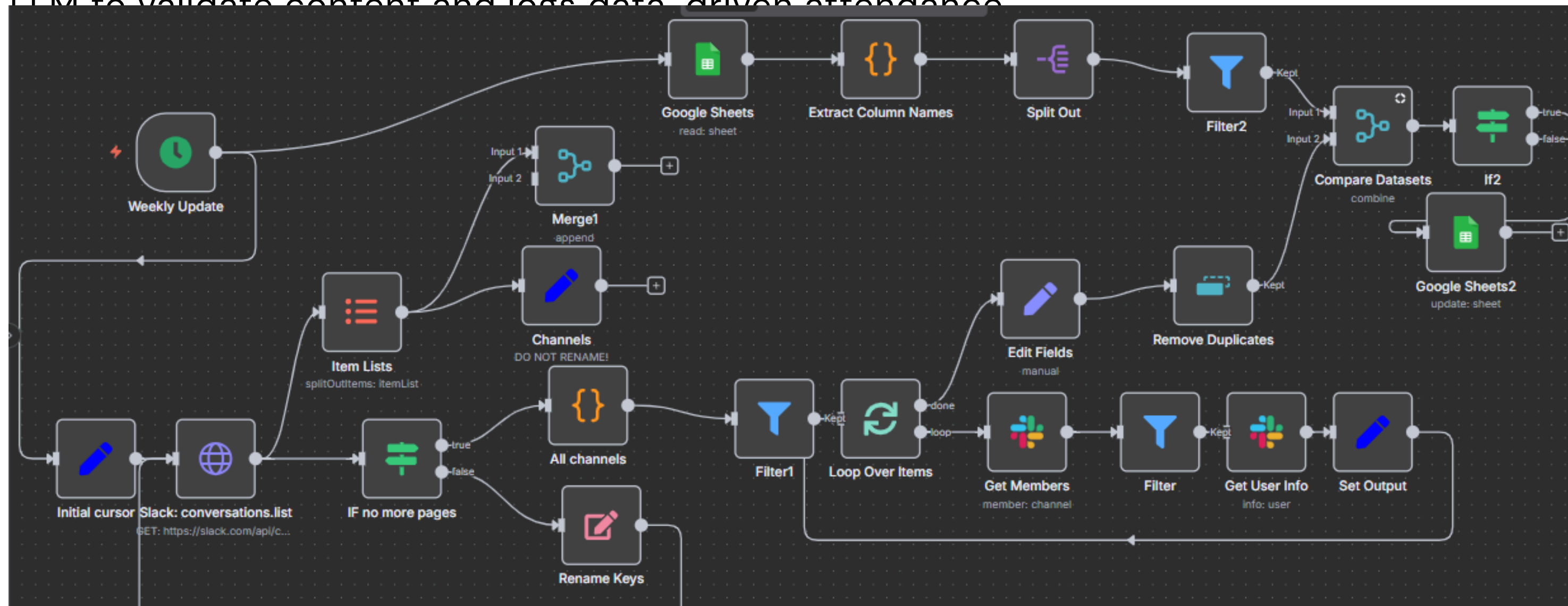
redis

Data Ingestion

IMPLEMENTATION

AGENTS

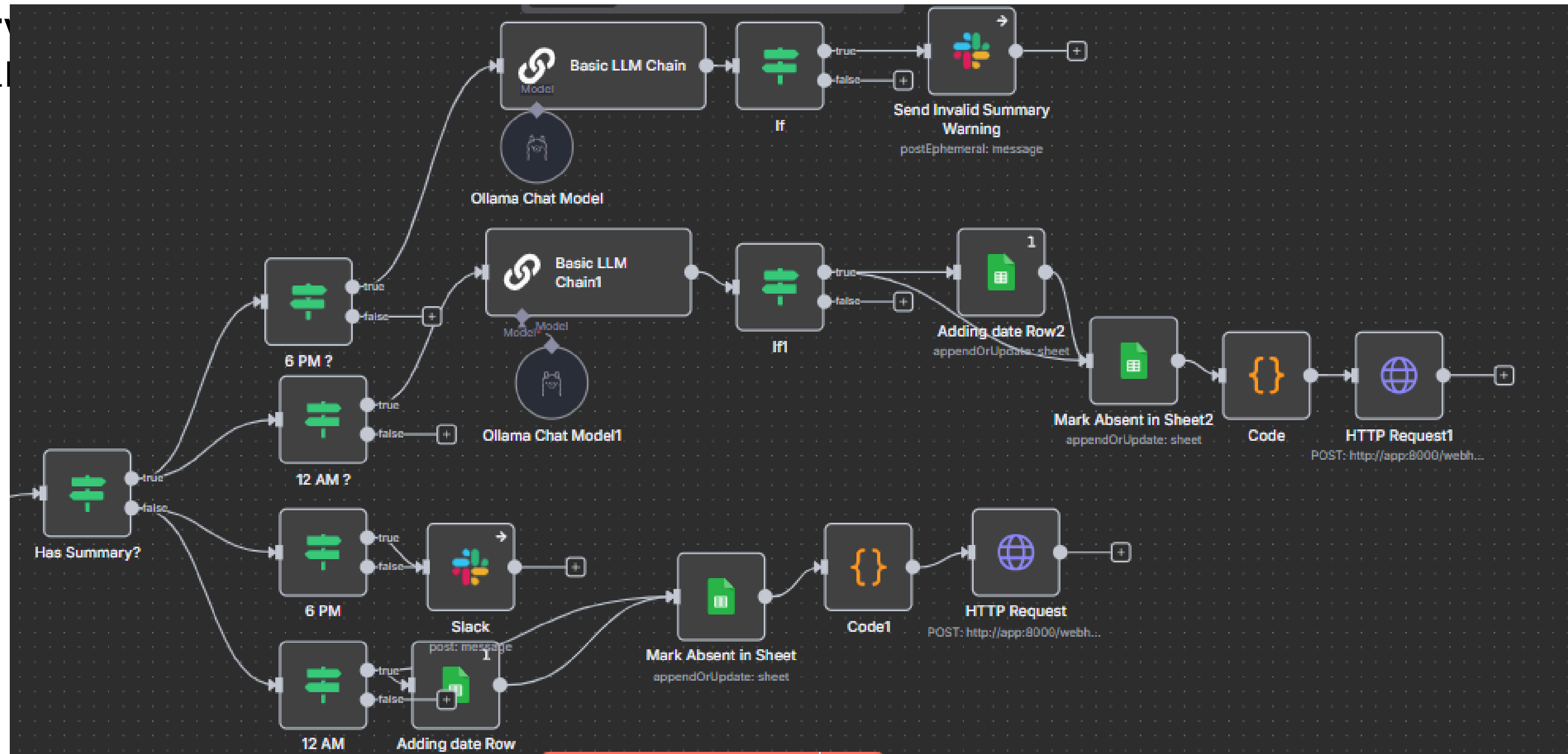
Supervision Agent (Slack): automates the supervision of daily team reports on Slack leveraging the LLM to validate content and log data-driven attendance.



AGENTS

Supervise the LLM

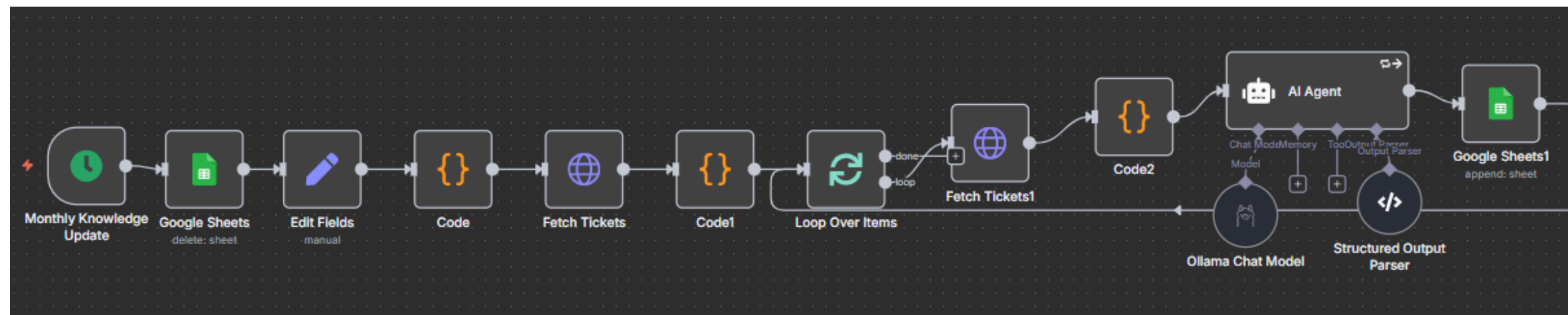
ng



AGENTS

Supervision Agent (Slack) : automates the supervision of daily team reports on Slack leveraging the LLM to validate content and logs data-driven attendance .

Support Agent (Zendesk) : ticket management agent that uses Retrieval-Augmented Generation (RAG) to instantly classify issues and draft factual, context-aware responses.



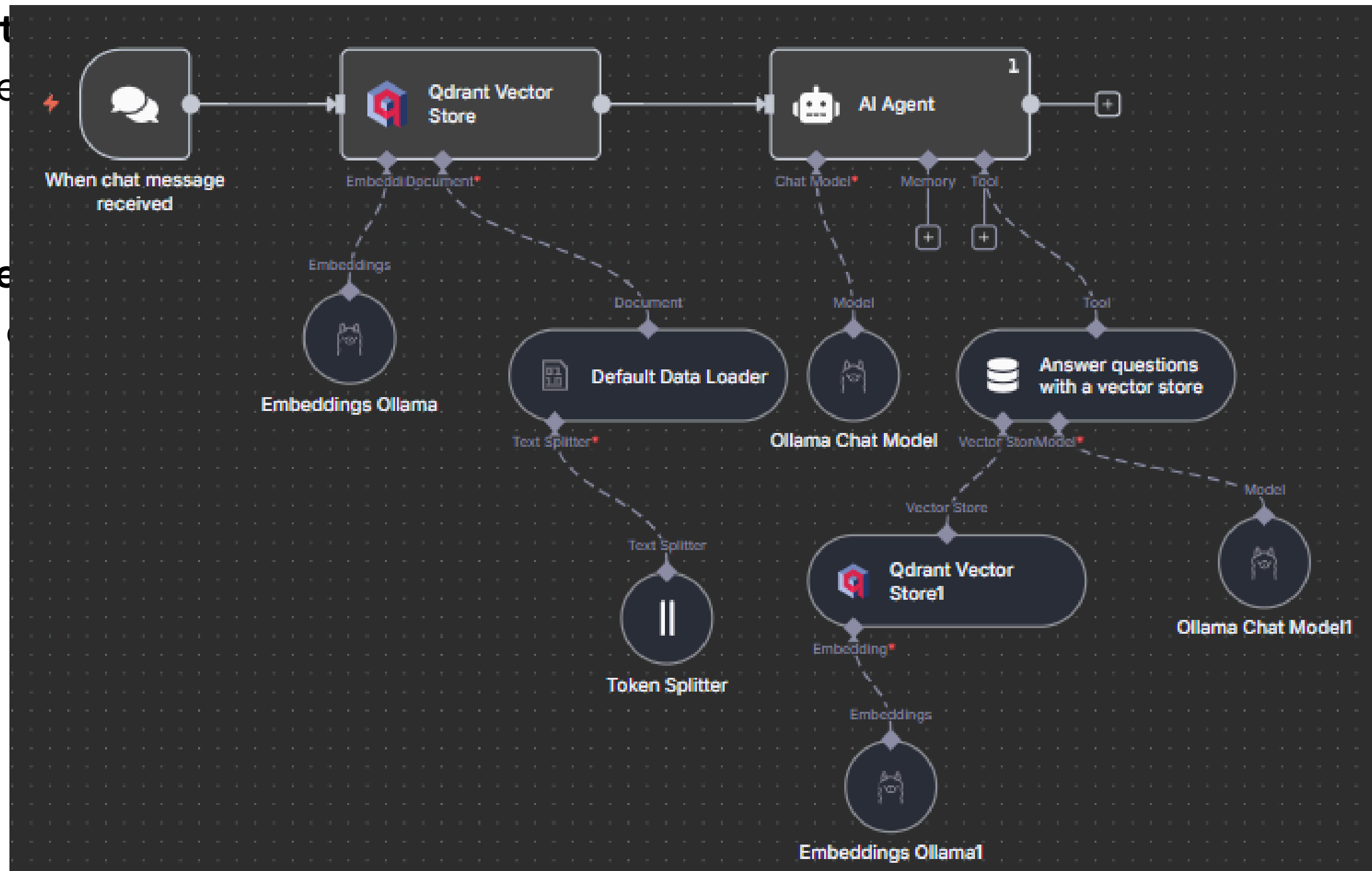
AGENTS

Supervision Agent
the LLM to validate

Support Agent (Zep)
(RAG) to instantly

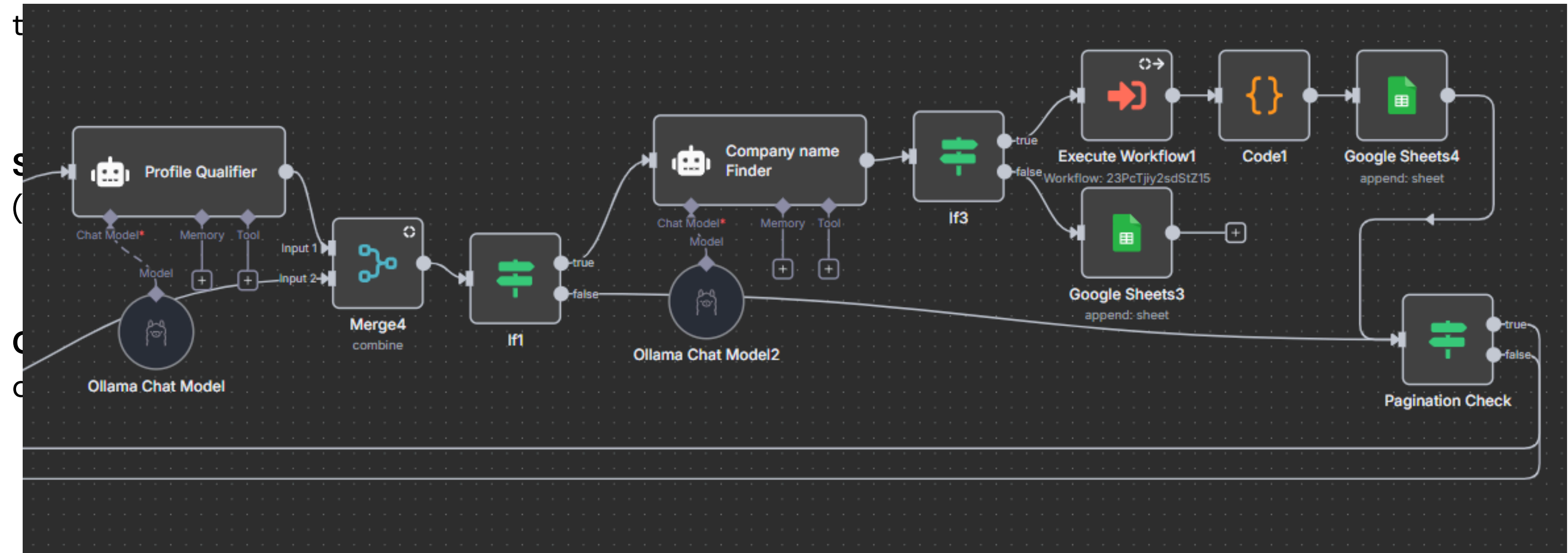
work leveraging

Generation



AGENTS

Supervision Agent (Slack): automates the supervision of daily team reports on Slack leveraging



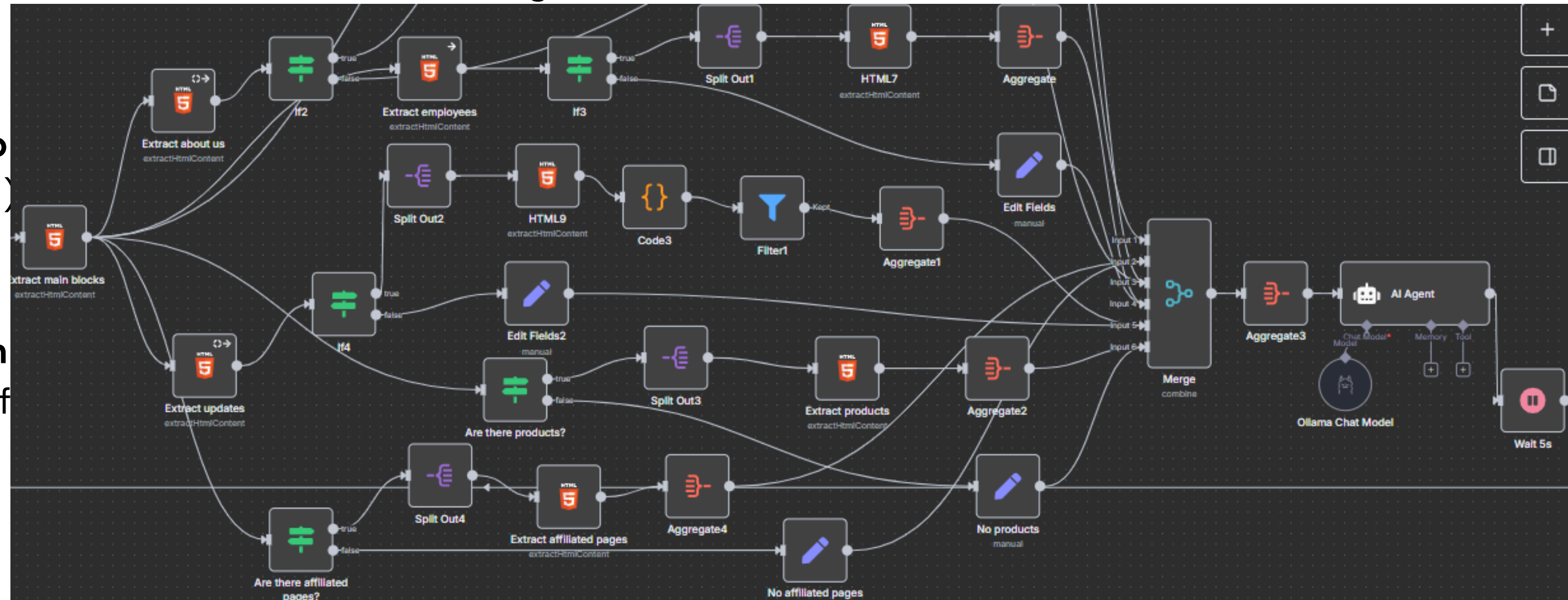
AGENTS

Supervision Agent (Slack): automates the supervision of daily team reports on Slack leveraging the LLM to validate content and logs data-driven attendance .

Supp
(RAG)

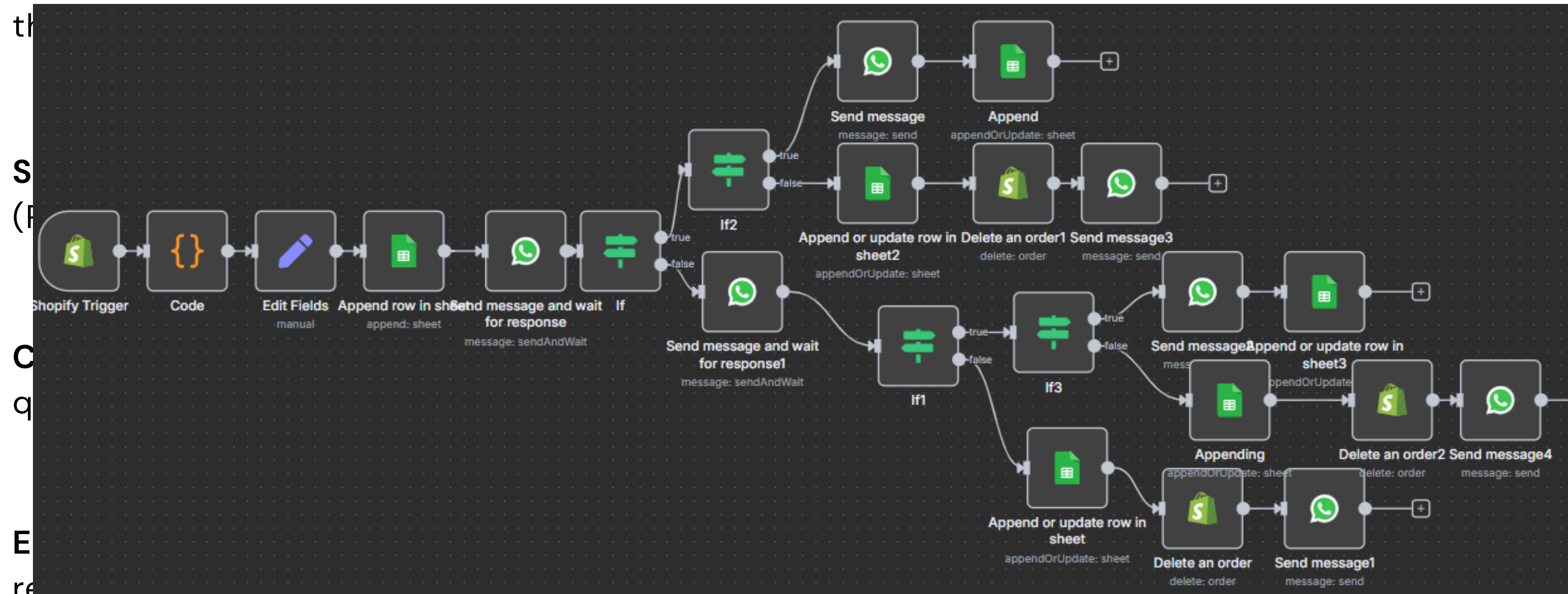
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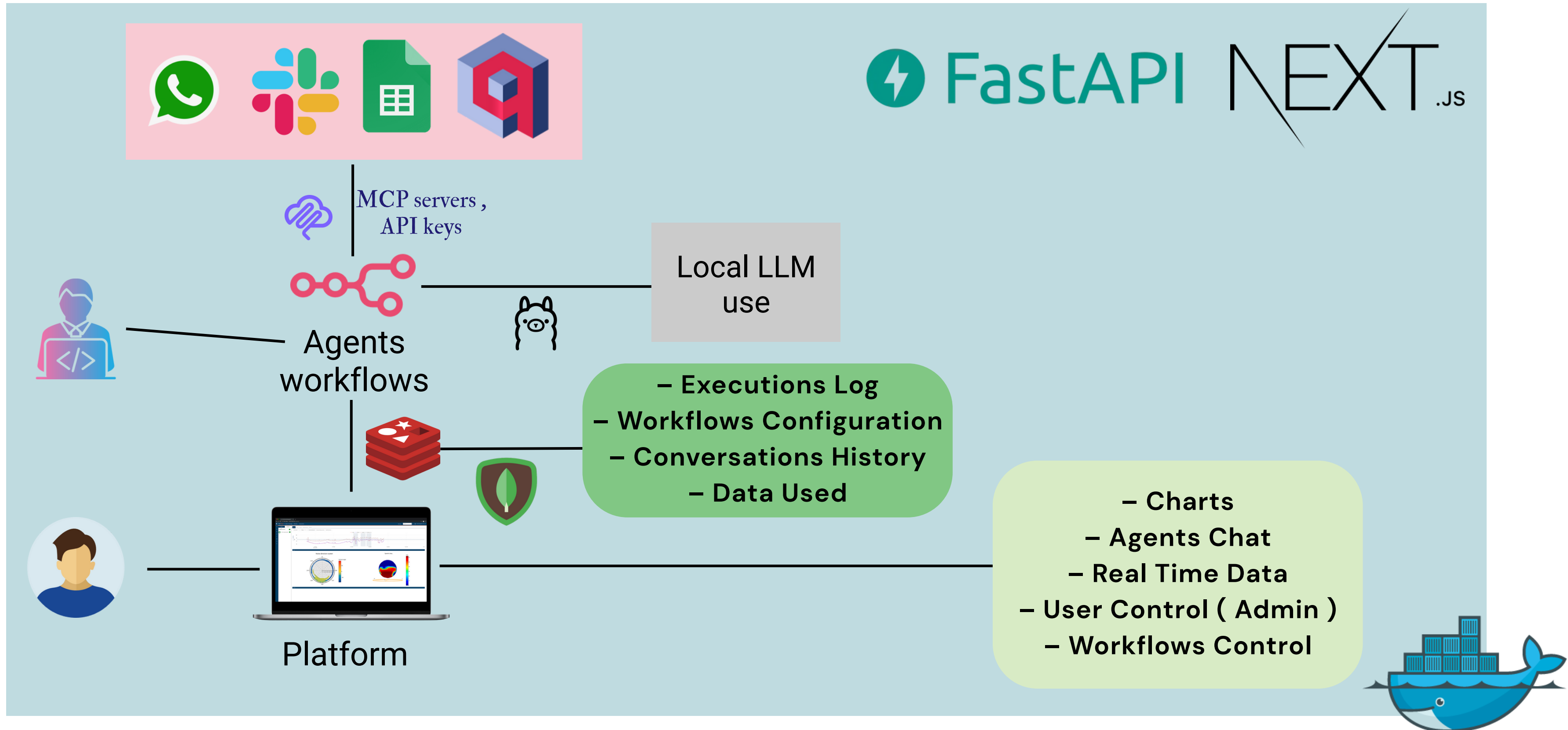
AGENTS

Supervision Agent (Slack): automates the supervision of daily team reports on Slack leveraging



SYSTEM Architecture

 FastAPI ~~NEXT~~.JS



SYSTEM

 OFFZONE AI SYSTEM

Dashboard

Admin Panel

Add New User

USERNAME	ROLE	WORKFLOW ACCESS	
admin@example.com	admin	None	Edit Delete
salaheddine.ouirra-etu@etu.univh2c.ma	user	Agent Commercial	Edit Delete
wireh65350@efpaper.com	admin	ZenDesk Agent	Edit Delete

Admin Dashboard

SYSTEM

Edit User: salaheddine.ouirra-etu@etu.univh2c.ma

Role

User

User

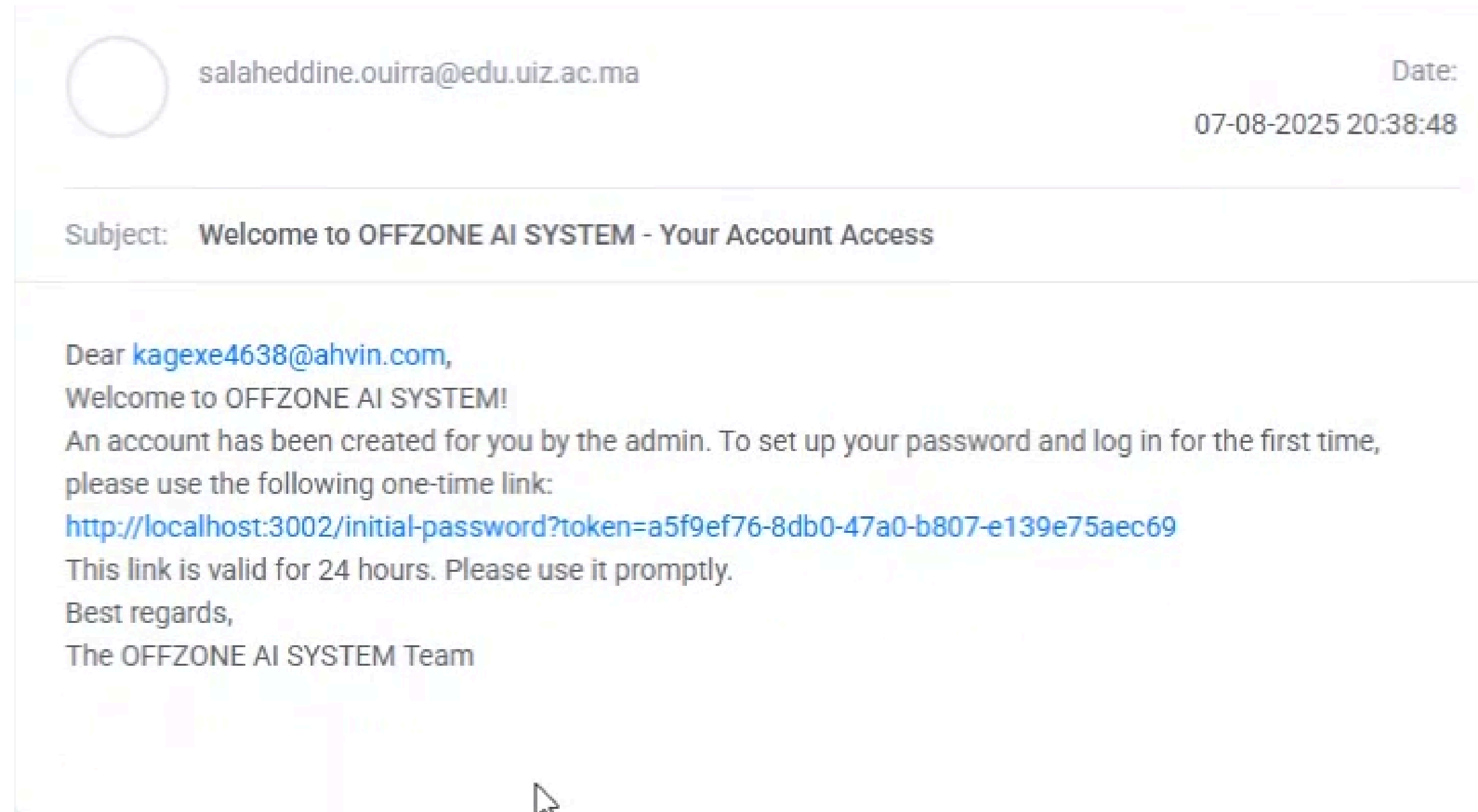
Admin

☐ n8n Law Guide	☐ Slack Agent With LLM
☐ ZenDesk Agent	☐ Extractor
☒ Agent Commercial	☐ DM Agent
☐ Slack Agent	☐ Sales Closer
☐ LogistiTrack	

Cancel Save Changes

Adding A User

SYSTEM



Email Verification

SYSTEM



Main Dashboard

SYSTEM

OFFZONE AI SYSTEM

Données - (Agent Commercial)

Execution ID: 1801
Status: success
Finished: Yes

Input:

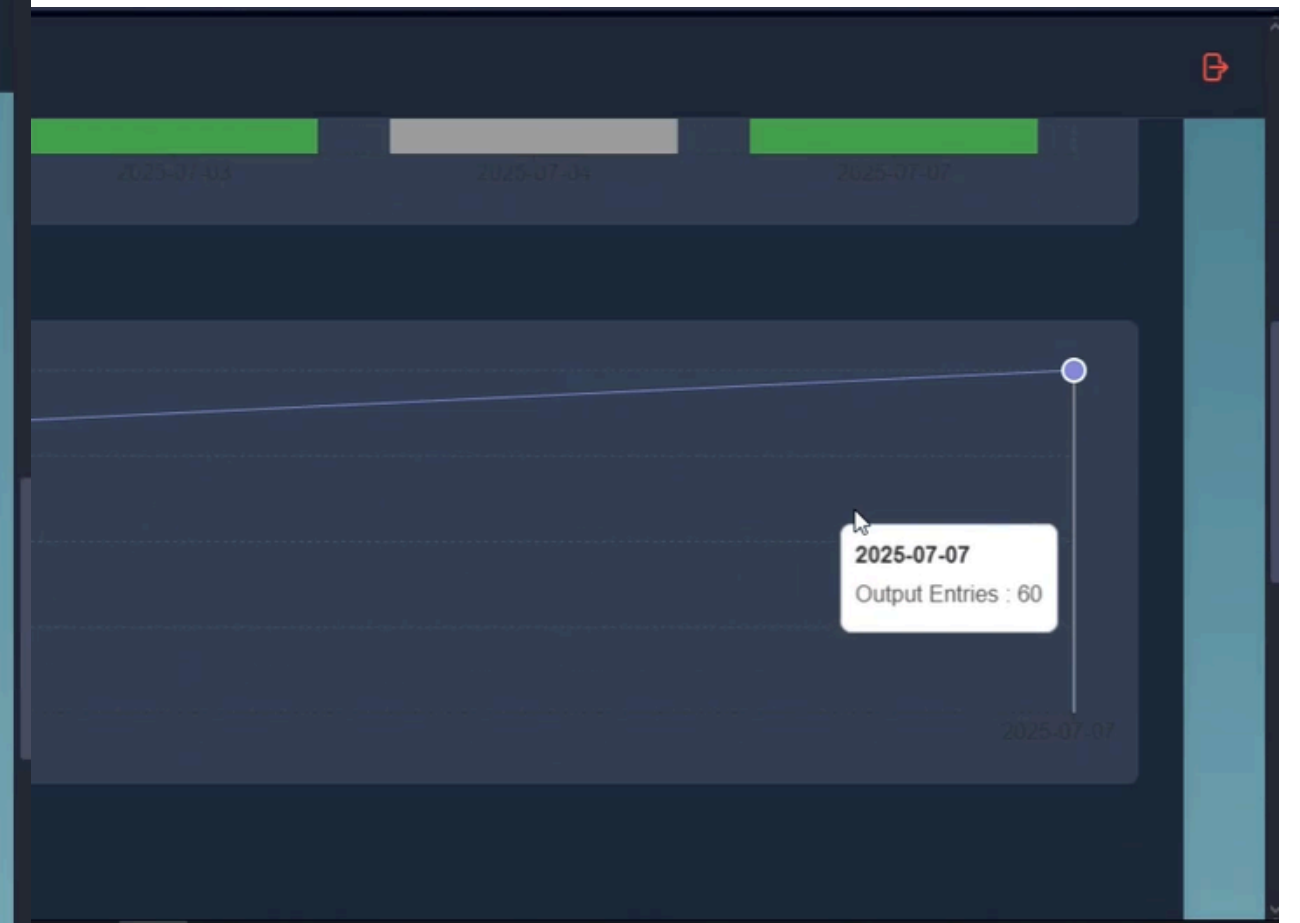
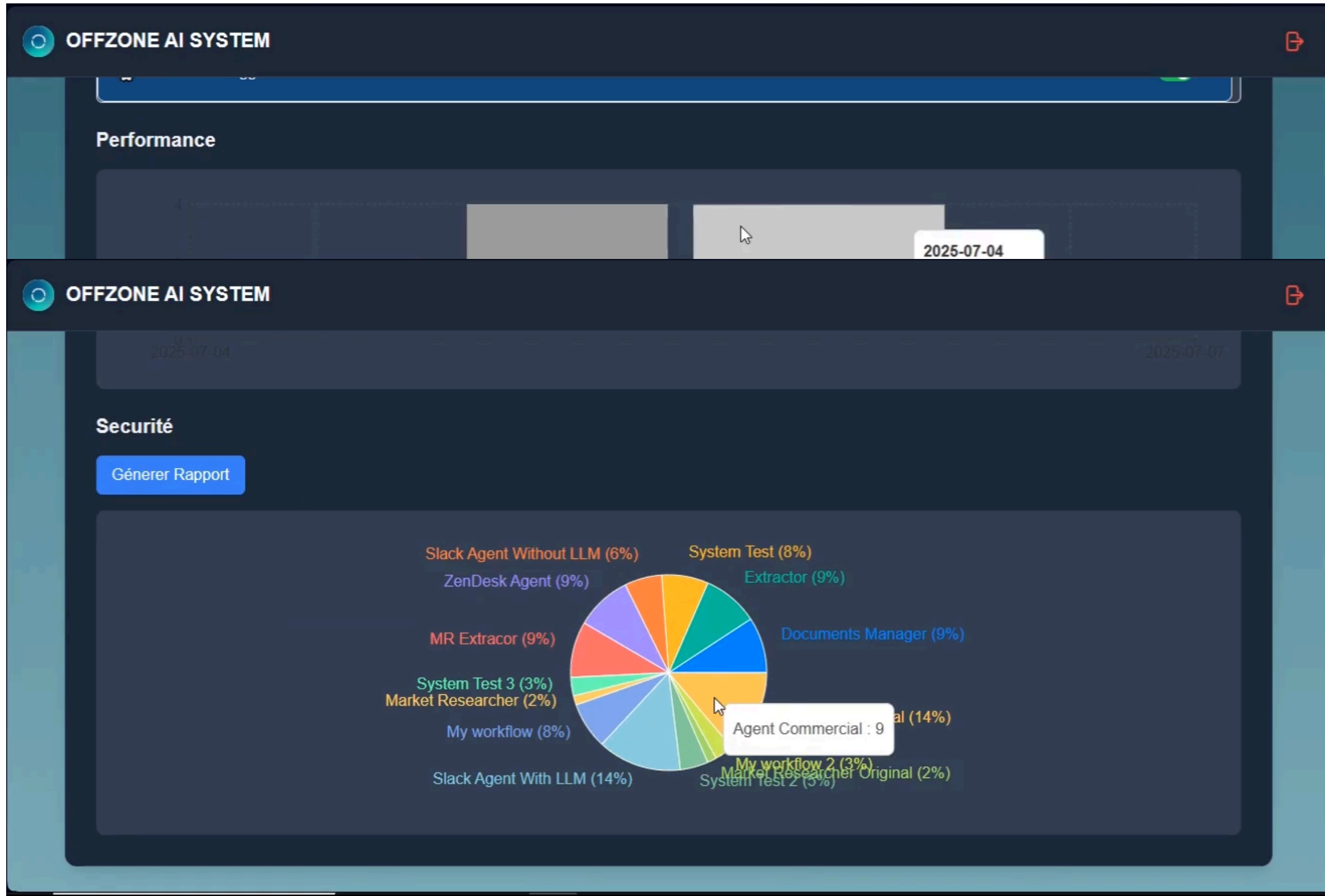
No input

Output Data:

IMAGE	INDEX	LINK	NAME	RESULTS	SNIPPET	TITLE
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQS_Pn6ebl-UxFPaTLZLJ4vyG_4PPgAMogd83l-5QohBvFdrPFd6DKf5fU&s	11	https://sg.linkedin.com/in/fbogacz	François Bogacz	true	... development to collective impact defines my approach to leadership development. · Experience: SMU-ExD Executive Development · Education: Telecom Paris ...	SMU-ExD Executive Development

Real Time Data

SYSTEM



Charts

SYSTEM

 OFFZONE AI SYSTEM

Dashboard

Agent Interaction

Your Agents

CA
Commercial Agent

DA
Data Analyst

DM
Documents Manager

MR
Market Researcher

ZA
ZenDesk Agent

SM
Slack Manager

CA Chat with Commercial Agent

Hello! How can I help you today?

chercher des CEOs en domaine IA au maroc

la recherche est commencé pour des CEO IA dans Maroc en excluant rien

Type your message...

Send

Chat with Agents

RESULTS

**Reduce Manual Task
Processing Time by 40–
60%**

**Achieve Over 95%
Compliance in Daily
Reporting**

**Decrease Average First-
Response Time by up to
40%**

**Generate a 3x to 5x
Higher Volume of
Qualified Leads**

**Reduce the Cash on
Delivery (COD) Return
Rate by 20–30%**

CONCLUSION



We have successfully moved beyond theory to build a functional and scalable multi-agent system.

By integrating a powerful LLM with a Retrieval-Augmented Generation architecture, our agents are not only automated but also accurate, context-aware, and reliable.





THANK YOU

I am at your disposal for any questions or comments.

Ouirra Salah Eddine

